# High-Speed Temperature Measurement During Polymer Deformation

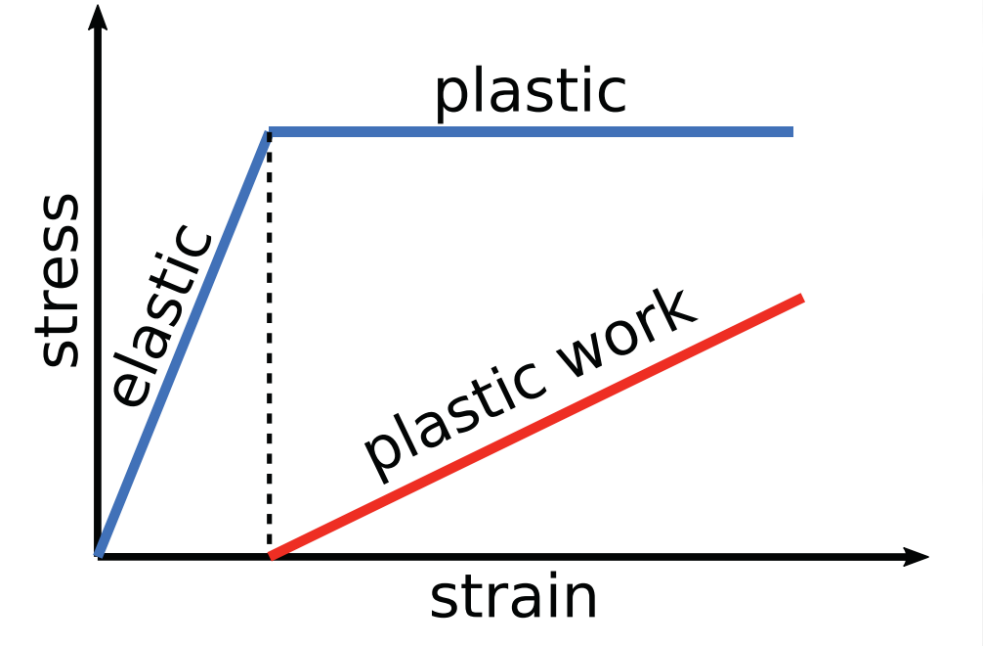
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Undergraduate Research Intern Summer 2026, Professor Amirkhizi

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# Polymer Deformation as a Thermal Event




$$E = Q + E_w$$

- $E$ → Plastic Work
- $Q$ → Heat
- $E_w$ → Stored Energy

Stimac, J. C., et al. "Energy Storage under High-Rate Compression of Single Crystal Tantalum." Acta Materialia, vol. 239, Oct. 2022, p. 118253. DOI.org (Crossref), https://doi.org/10.1016/j.actamat.2022.118253.

# Conversion of Plastic Work to Heat

$$\beta_{\mathrm{diff}} = \frac{\rho c \dot{T}}{\mathrm{d}\dot{W}_{\mathrm{p}}}$$

- Instantaneous Conversion

$$\beta_{\mathrm{int}} = \frac{\rho c\, \Delta T}{\int \mathrm{d}\, W_{\mathrm{p}}}$$

- Cumulative Conversion

# Necessary Temporal Resolution

IR Camera (FLIR A655sc):

- Spatial distribution
- Locate hot regions
- Full thermal field




- IR Camera has slower temporal sampling
- IR Sensor only measures one controlled region

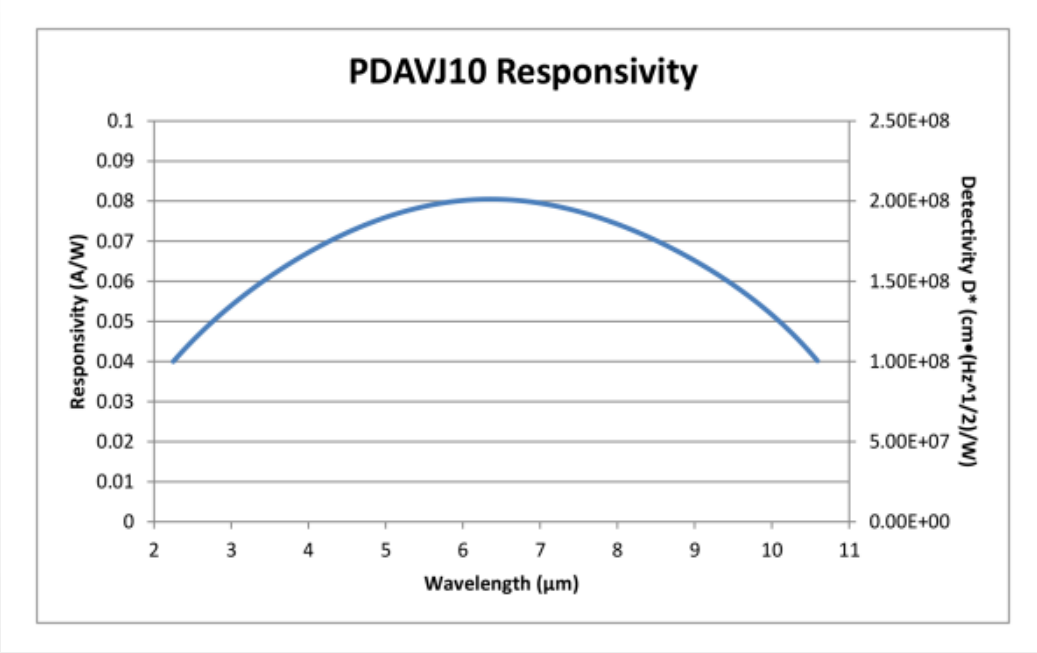
IR Sensor (Thorlabs PDAVJ10):

- High temporal response
- Continuous voltage output



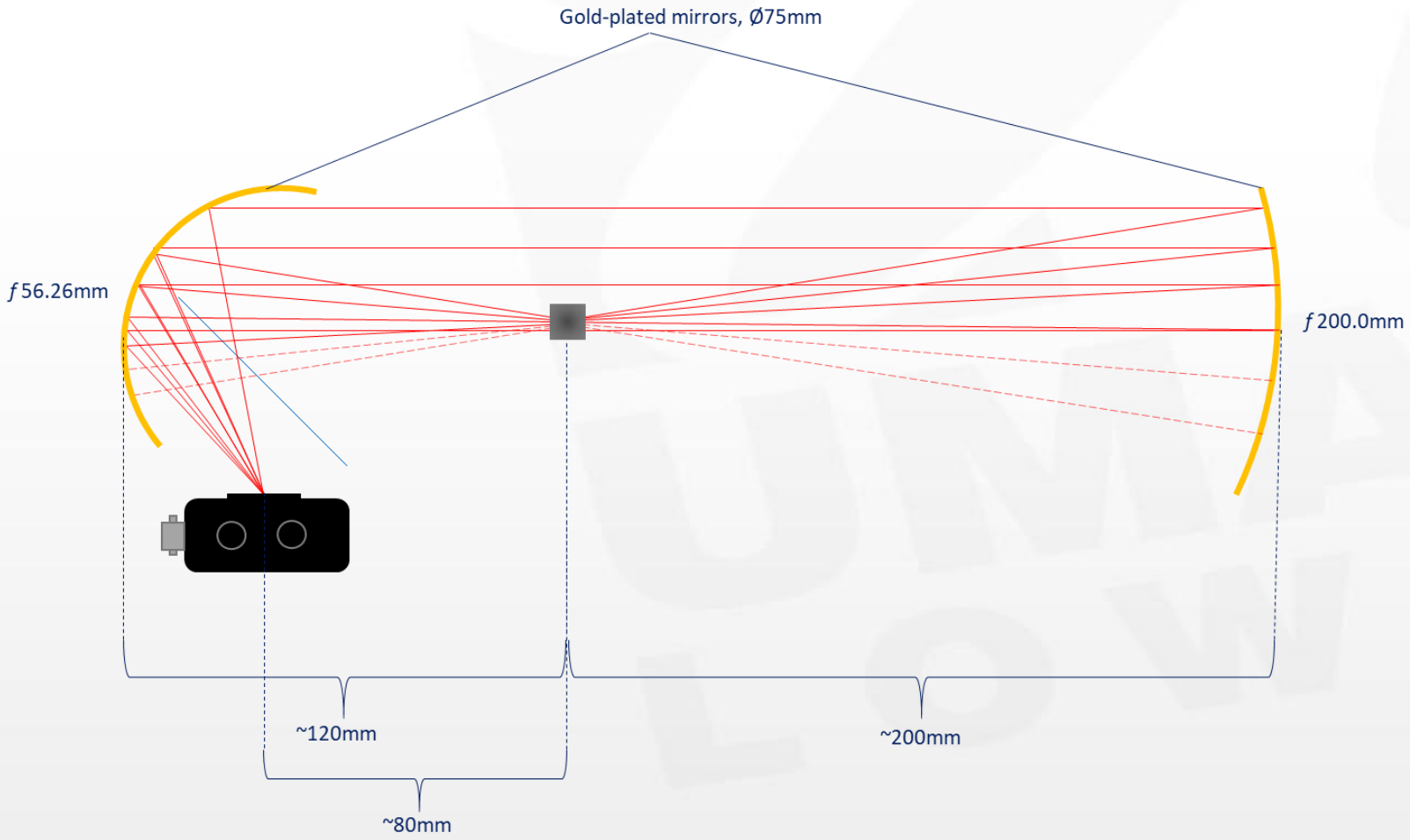
# High-Speed Infrared Detection

- Thorlabs PDAVJ10 HgCdTe detector
- 1mm$^2$ effective active area
- Built-in amplifier

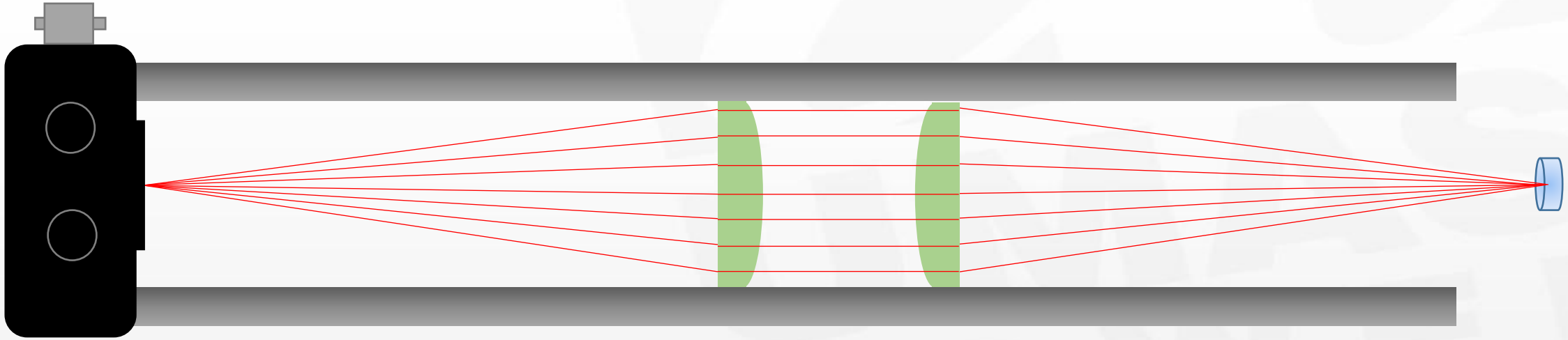




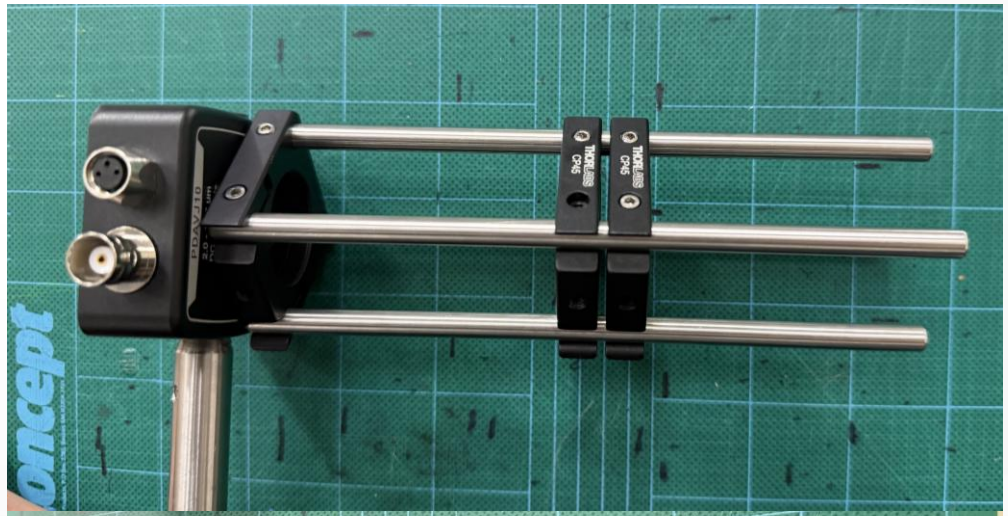

# Original Optical System

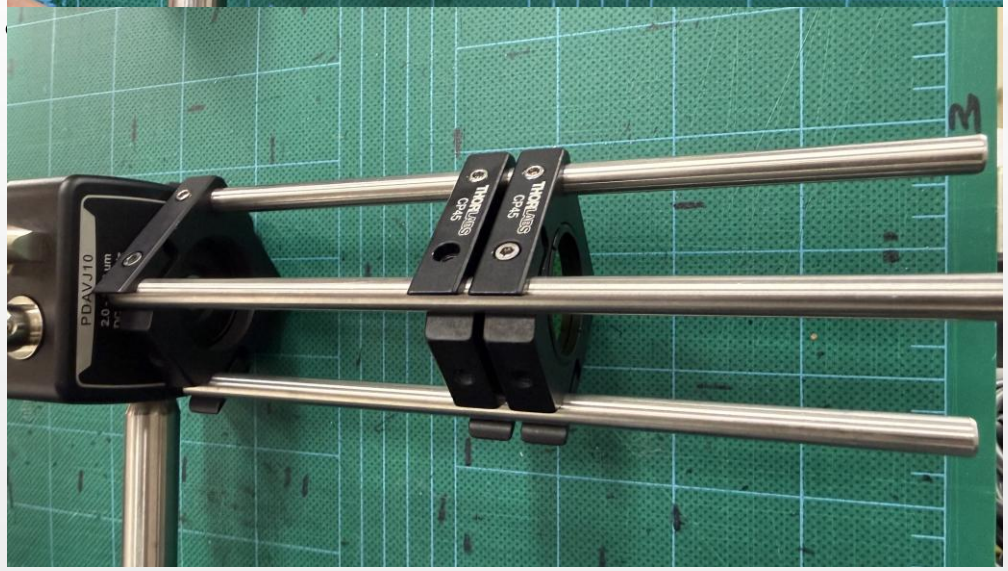


# Optical Redesign



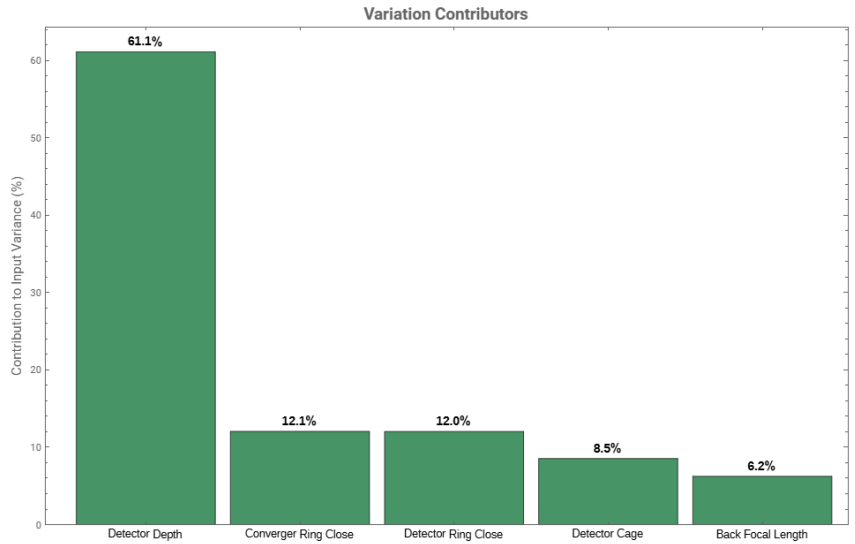
# Precise Assembly of Optical Setup



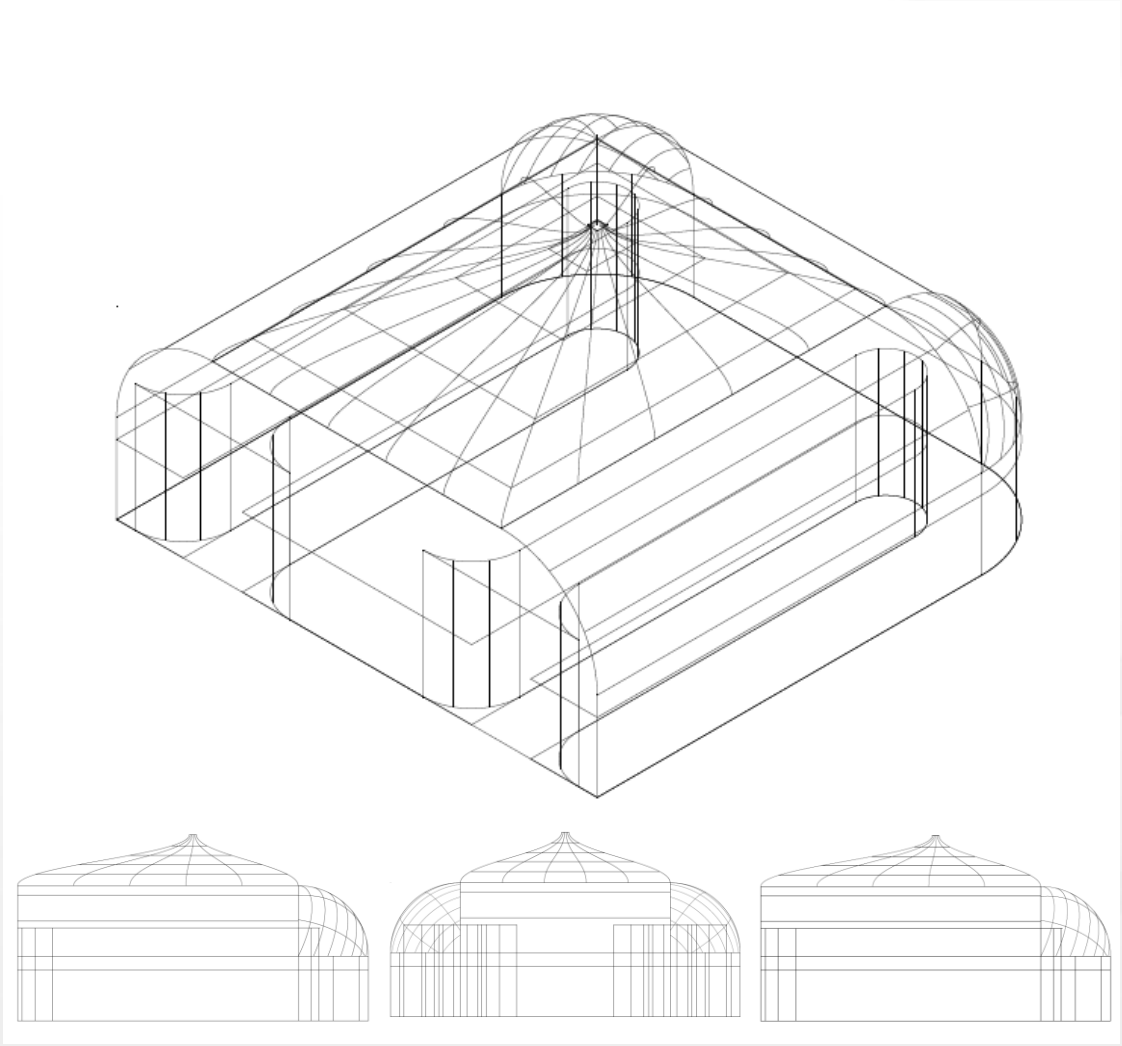

**Model Assumptions**

| Input | Model | Range |
|---|---|---|
| Back Focal Length | Uniform | 75.17 - 75.33 |
| Detector → Ring Depth | Uniform | 4.07 - 4.57 |
| Detector Ring Close | 50/50 Uniform Mixture | 1.87-2.13 / 2.05-2.09 |
| Detector Cage | 50/50 Uniform Mixture | 8.77-9.03 / 8.87-8.91 |
| Converger Ring Close | 50/50 Uniform Mixture | 1.87-2.13 / 2.05-2.09 |



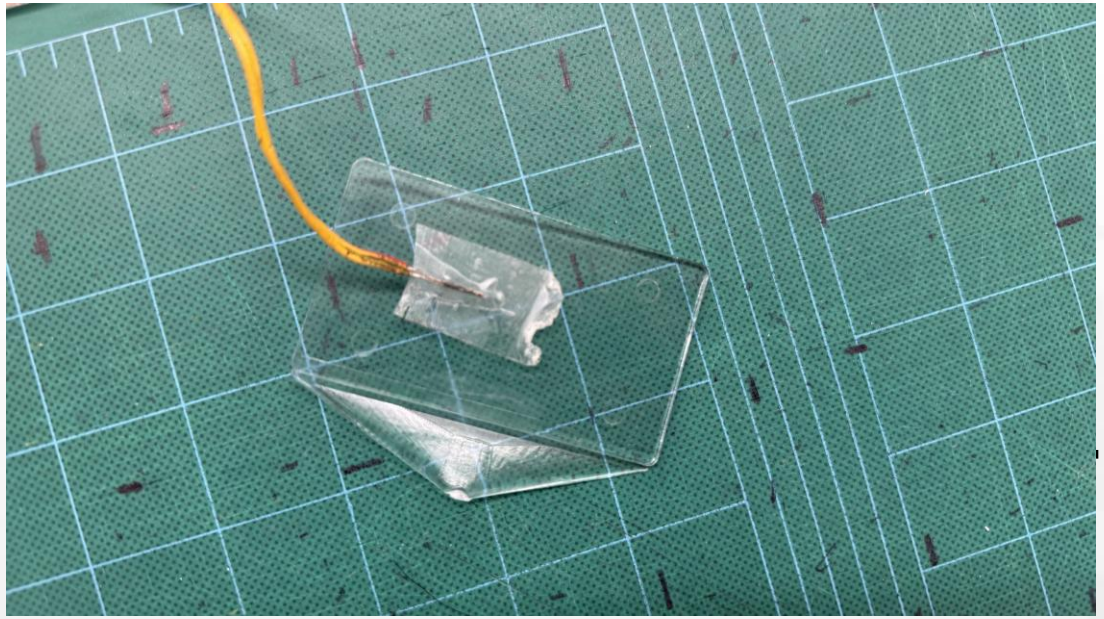

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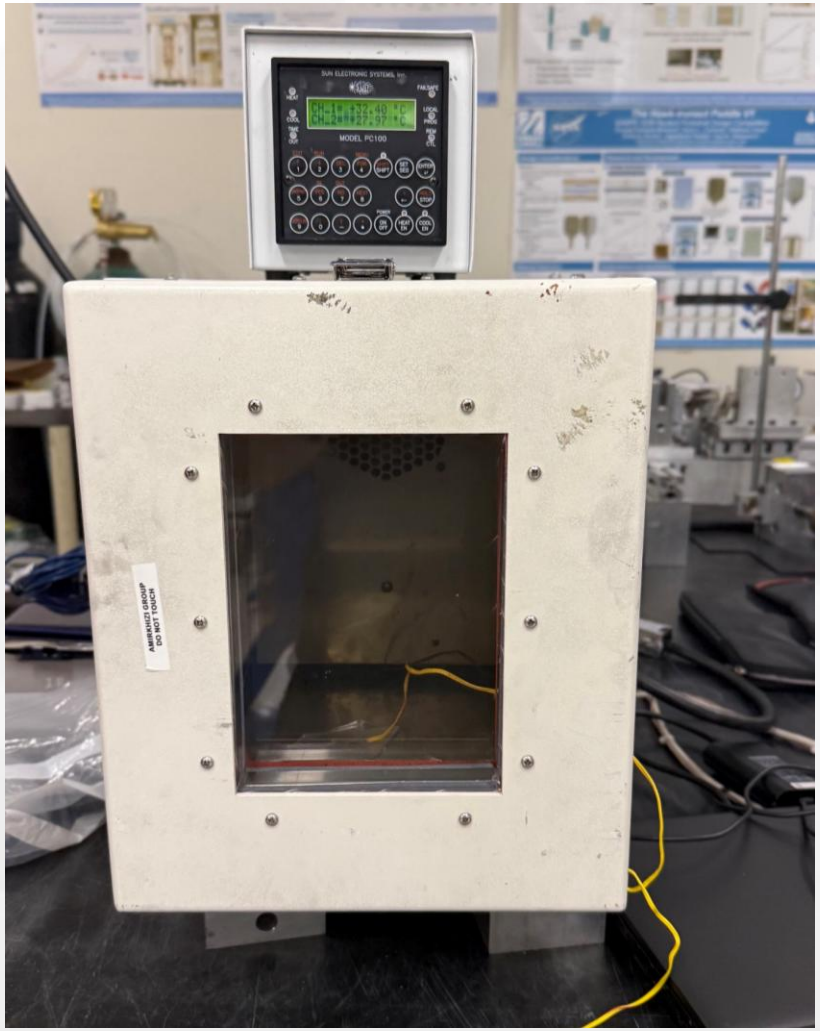
# Optical Alignment



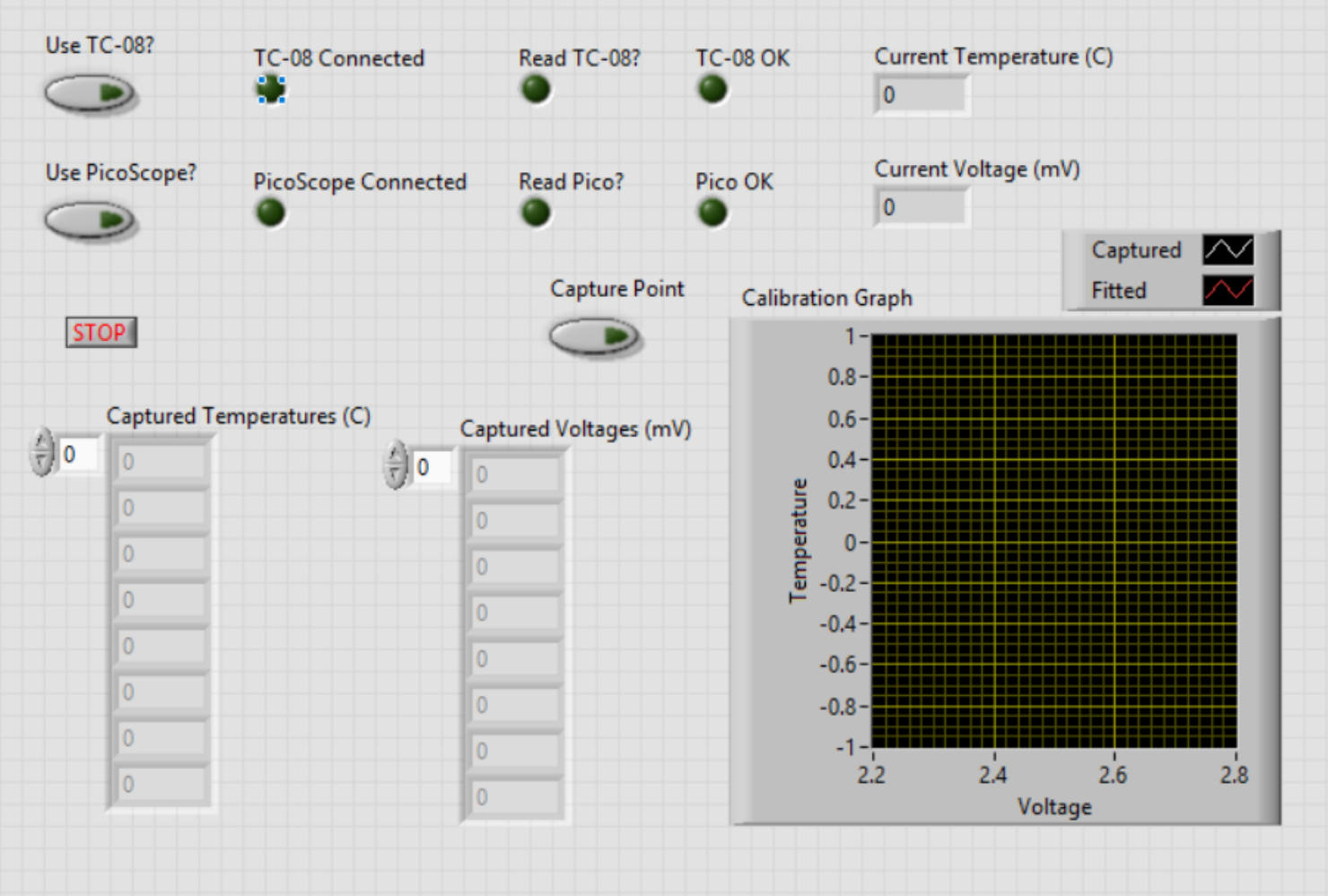


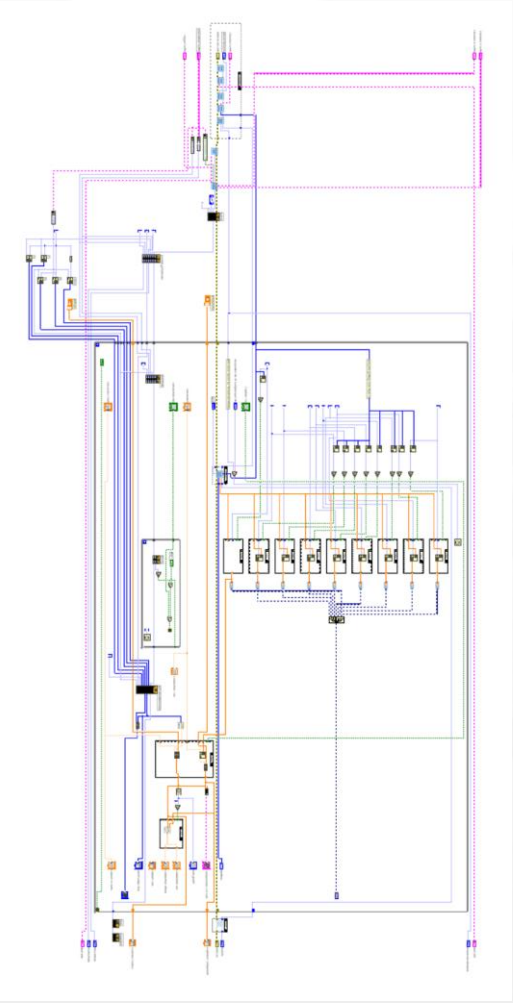
# Sensor Calibration



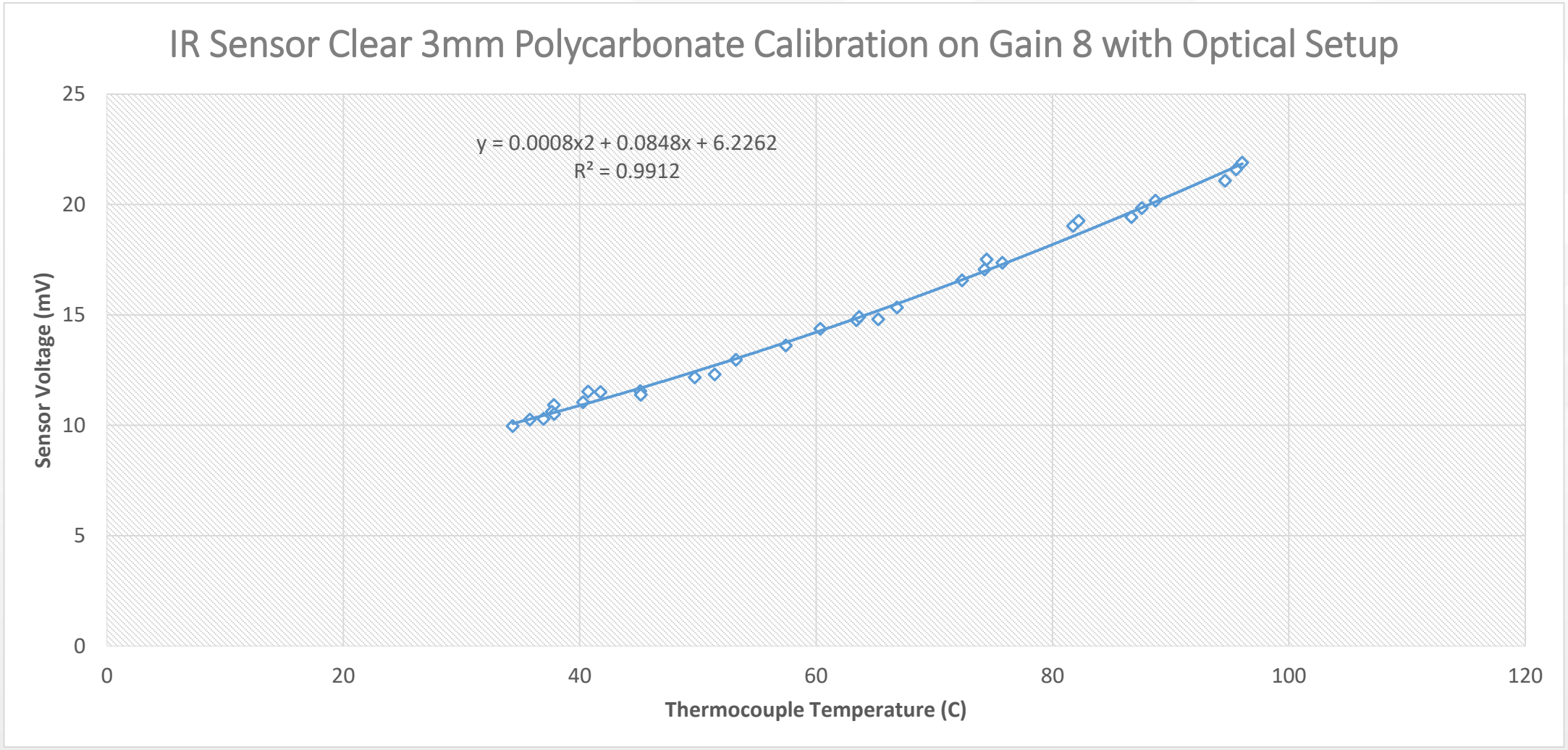


# Synchronized Data Acquisition

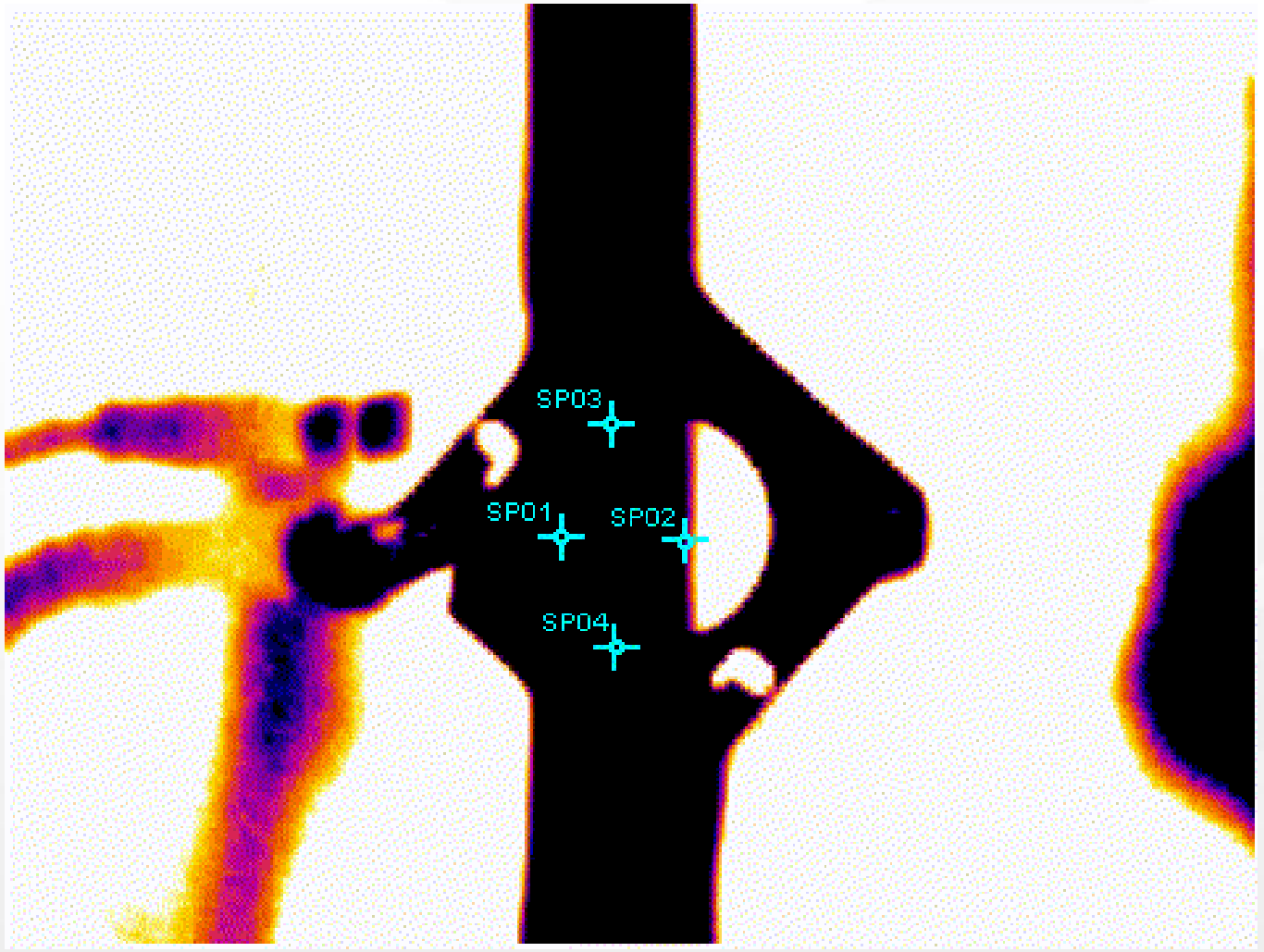


# Calibration Results



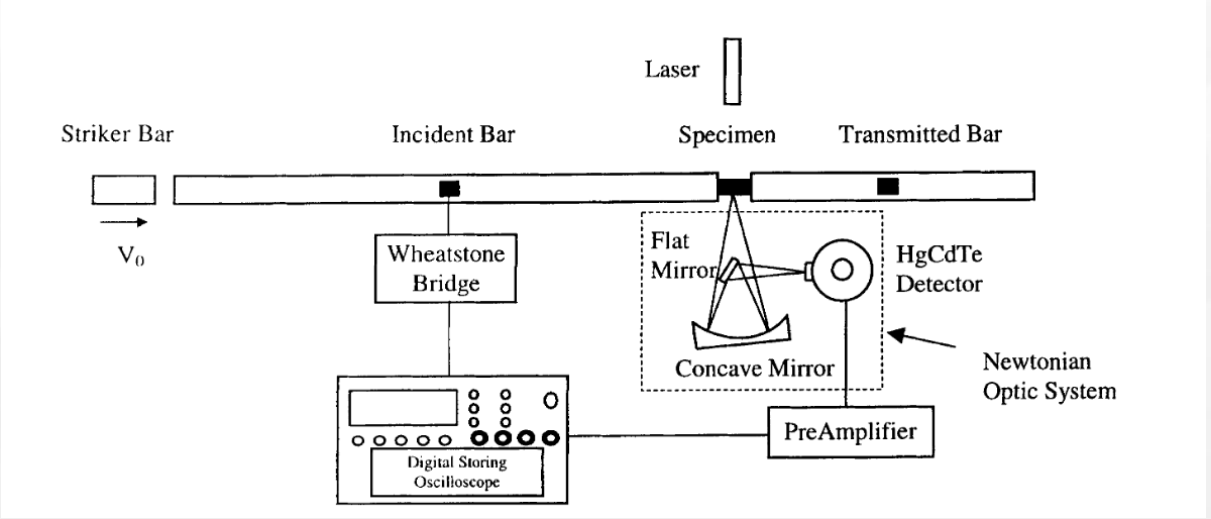
# Quasi-Static Tensile Validation



# Quasi-Static Tensile Validation

# Extension to High-Strain-Rate Testing






Li, Zhouhua, and John Lambros. "Strain Rate Effects on the Thermomechanical Behavior of Polymers." *International Journal of Solids and Structures*, vol. 38, no. 20, May 2001, pp. 3549–62. *DOI.org (Crossref)*, https://doi.org/10.1016/S0020-7683(00)00223-7.

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# Questions?